

Inequality and Climate Change Impacts

Does Economic Inequality Increase Country Emissions?

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Research Puzzle

Motivating Question

- Climate Change is an existential global challenge
- Countries and people need to reduce their emissions *yesterday*
- Economic inequality leads to elites having more political power
- National climate policies have a large impact on emissions

Me: Drinking from a soggy straw to save the planet

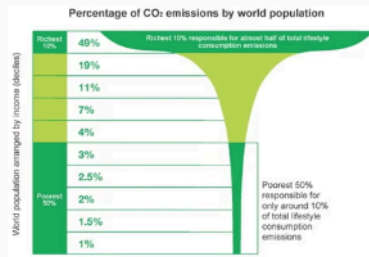
Taylor Swift:



Research Question

Motivating Question

- Emissions are unequal within countries
 - Oxfam: The world's richest 1% are responsible for more carbon emissions than the poorest 66%
- **Do more unequal countries do less to mitigate climate change?**



What we know

- There is wide variation in global emissions
- Much is driven by wealth
 - Wealth -> Consumption -> Emissions
- Generally, people are supportive of climate policies, but pushback from rich elites can hinder climate action

I derive the following expectations:

- **H1 (Main):** Ceterus paribus, as *Income inequality* increases, *Per Capita carbon emissions* will also increase

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- **H1 (Main):** *Ceterus paribus*, as *Income inequality* increases, *Per Capita carbon emissions* will also increase
- **Null Hypothesis** There is no relationship between income inequality and carbon emissions.

Income Inequality - Various Measures

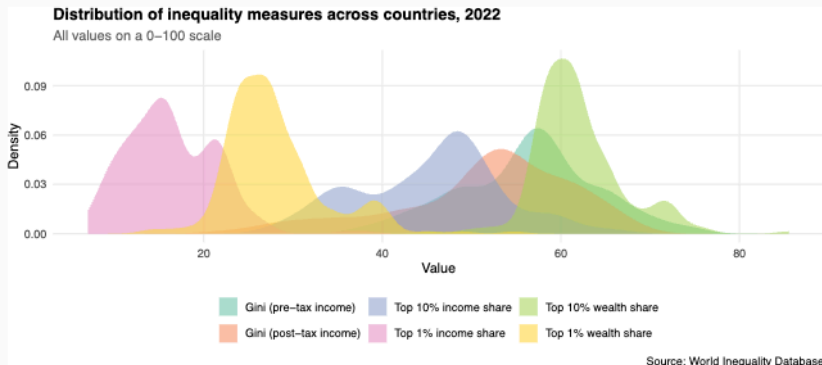


Figure 1

Primary Independent Variable

- Primary IV: Income Inequality - The distribution of wealth within a country
 - Operationalized using the share of income to the top 10% within each country
 - Date from the World Income Database

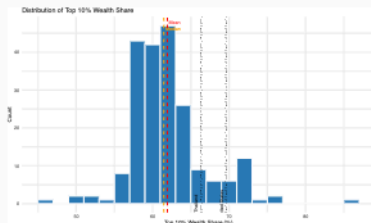
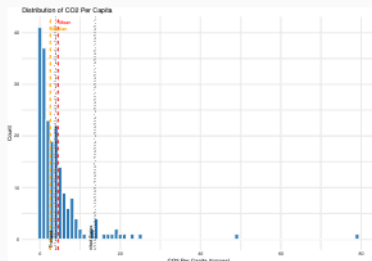


Figure 2

Dependent Variable

Carbon Emissions - The per capita contribution to climate change - Operationalized as CO2 Emissions in metric tons per capita - Data from the world bank

All data are from 2023 or most recent



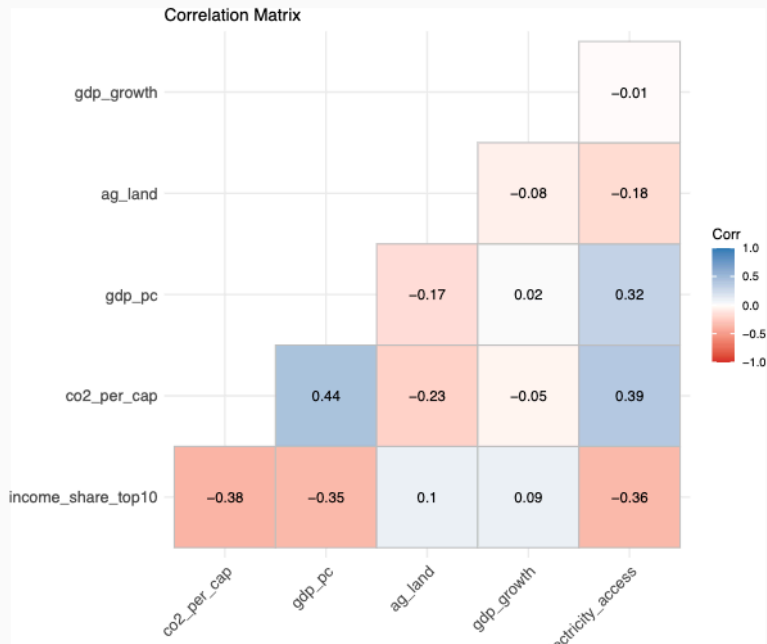
Descriptive Statistics and Controls

Table 1: Summary Statistics

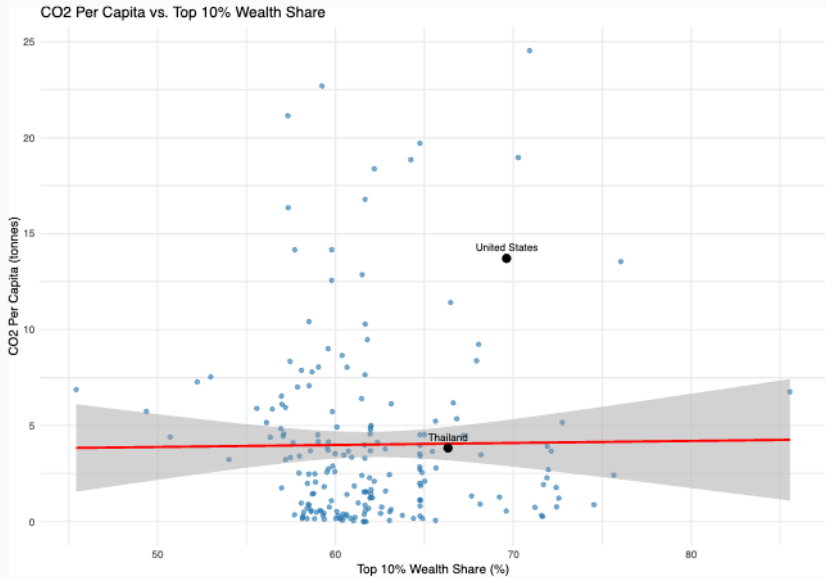
Table 2: Table 1: Descriptive Statistics

Variable	Mean	Median	SD	Min	Max	N
Income Share: Top 10%	28.41	27.45	5.16	19.90	42.70	146
CO2 per Capita (tonnes)	3.91	2.49	4.66	0.00	24.53	201
GDP per Capita (thousands USD)	22.04	7.82	33.47	0.25	256.80	202
Agricultural Land (%)	37.44	38.12	22.57	0.45	86.48	207
GDP Growth (%)	3.42	3.20	6.84	-29.43	75.31	202
Electricity Access (%)	88.07	100.00	22.51	5.40	100.00	213

Correlation Matrix



Initial Scatter Plot



Primary Regression Results

Table 3: OLS Regression — Main Results

	Bivariate	+ GDP Per Capta	Full Model	Add Controls
(Intercept)	10.752*** (1.534)	5.812*** (1.454)	6.325*** (1.533)	1.372 (1.910)
income_share_top10	−0.255*** (0.053)	−0.129*** (0.048)	−0.121** (0.048)	−0.075 (0.047)
gdp_pc		0.079*** (0.010)	0.079*** (0.011)	0.069*** (0.010)
ag_land			−0.020* (0.011)	−0.016 (0.011)
gdp_growth			0.033 (0.085)	0.067 (0.081)
electricity_access				0.041*** (0.010)
Num.Obs.	141	140	140	140
R2	0.143	0.411	0.425	0.487
R2 Adj.	0.136	0.402	0.408	0.467

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

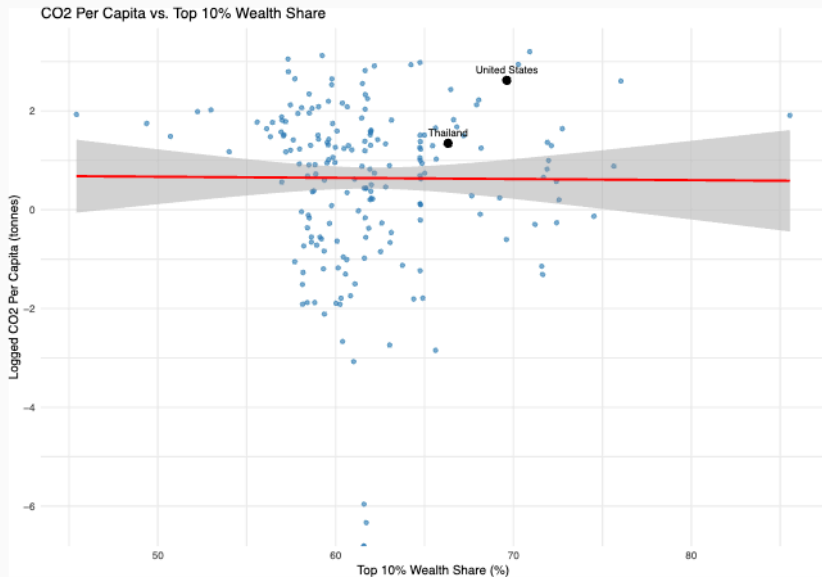
Logged Regression Results

Logging GDP per Capita and CO2 per Capita

	Bivariate	+ GDP Per Capita	+ Land & Growth	+ Electricity
(Intercept)	3.369*** (0.635)	-0.440 (0.493)	-0.796 (0.529)	-2.205*** (0.570)
Income Share Top 10%	-0.098*** (0.022)	-0.017 (0.015)	-0.014 (0.015)	-0.007 (0.014)
ln GDP per Capita		0.776*** (0.055)	0.830*** (0.059)	0.595*** (0.073)
Agricultural Land			-0.002 (0.004)	-0.002 (0.003)
GDP Growth			0.072*** (0.027)	0.061** (0.025)
Electricity Access				0.020*** (0.004)
Num.Obs.	141	140	140	140
R2	0.126	0.647	0.665	0.714
R2 Adj.	0.120	0.642	0.655	0.704

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Plotting with CO2 Logged



Standardized Regression

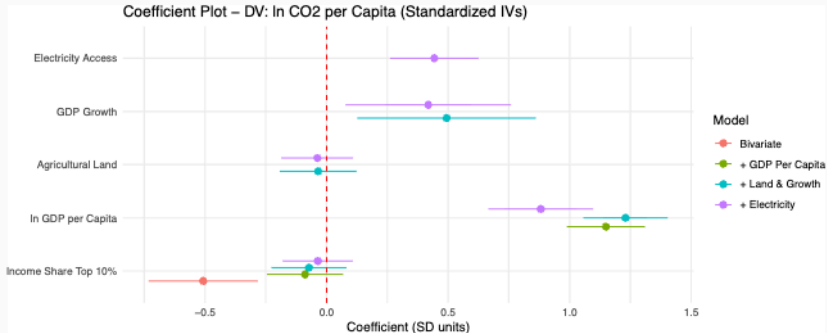


Figure 3: Coefficient Plot

Alternative Measures of Wealth Inequality

Includes all control variables

	Income Share Top 10	Gini (post-tax)	Gini (pre-tax)	Income Share Top 1	Wealth Top 10	Wealth Top 1
Income Share Top 10%	-0.007 (0.014)					
Gini (post-tax)		0.015* (0.008)				
Gini (pre-tax)			0.012 (0.010)			
Income Share Top 1%				0.016 (0.014)		
Wealth Share Top 10%					0.018 (0.014)	
Wealth Share Top 1%						0.017 (0.012)
Num.Obs.	140	186	186	186	186	186
R2	0.714	0.633	0.630	0.629	0.630	0.631
R2 Adj.	0.704	0.623	0.619	0.619	0.620	0.621

* p < 0.1, ** p < 0.05, *** p < 0.01

There does not appear to be a strong relationship between wealth inequality and CO2 Emissions

- Potential Omitted Variable Bias
 - Political Institutions - Democracies and Interest group systems - ie how elites preferences are heard
 - Oil Dependent Economies
 - Regional Differences
- Could look at climate policy as DV, rather than emissions

Qualitative Follow-Up

To probe the mechanism, future work should conduct process tracing to better understand if or how economic elites shape climate policy.

- United States as a most crucial case
- Thailand as “typical” case

Case studies of wealthy people/companies in each country

Policy Implications

These results suggest wealth inequality is not a meaningful impediment to effective climate policy

Limitations

- One year snapshot
- Country Level of analysis
- Aggrgate Numbers

Thank You

I look forward to your questions

Slides code available on course website